

AN ECLECTIC TECHNOLOGIES FOUNDATIONAL DEFINITION

FD-BL-D1 — Mode of Discharge: Pre-Authorization by Verified Gatekeeper

Version 0.3 — DRAFT · August 2026

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Designation: FD-BL-D1 **Class:** Determination under FD-BL (not a Foundational Definition — see §2) **Governing instrument:** FD-BL — Foundational Definition — The Bright Line, v0.2, §6.1 **Version:** 0.3 — DRAFT (§9 terminology bullet conformed to FD-EV; supersedes v0.2) **Date:** August 2026 **Author:** Tim Roxey **Organization:** Eclectic Technologies **Companion constructions:** CB Proposer–Gate Pattern; DBA-VC Vendor Companion (consequence classifier); DBA-GAS-FC2 (operating-envelope-with-setback)

1. Purpose

FD-BL §4.5 requires that an action placed above the Bright Line have substantive, attributable human authorization by a qualified Command Broker, enforced architecturally, and that the requirement is not discharged by a procedure instructing an operator to confirm. FD-BL §6.1 leaves one question open: whether that authorization may, for a narrow and specified class of actions, be discharged **in advance** rather than in the moment. This instrument determines that question.

This determination resolves FD-BL §6.1. On issue, FD-BL §6.1 is reduced to a pointer — “*the mode-of-discharge question is resolved; see FD-BL-D1*” — and this instrument carries the resolution and its amendment set.

2. Authority, class, and what this does not alter

2.1 Class

FD-BL-D1 is a **determination**, not a foundational definition. By the family test at FD-BL §2.2, a concept belongs to the Foundational Definitions family only when the correctness of other documents depends on agreeing what *kind of thing* it is. This instrument does not define a kind of thing; it resolves the **mode** by which an existing obligation may be discharged. It is therefore designated as a determination bound to FD-BL rather than as a peer of FD-BL or FD-LD.

2.2 What this determination may and may not do

This determination operates under the constraint FD-BL §6.1 places on it, restated here as binding:

It varies the mode of discharge of the §4.5 authorization. It does not vary placement (FD-BL §4.2), granularity (FD-BL §4.3), or scope (FD-BL §4.4). An action eligible for advance discharge under this

instrument remains an above-line action, placed on the consequence of the individual action, produced

by a component within scope of the obligation. Nothing here reclassifies any action to below the line, and nothing here creates an intermediate class of action.

2.3 The governing principle it applies

Discharge is a verification question, governed by FD-BL §4.5: the Command Broker is compensation for unverifiability, and where an authorization decision is itself **bounded by proof** the compensation is not required. This is the same principle by which a deterministic protective function performs the highest-consequence actions without a broker (FD-BL §4.5): what it will do is known before it does it, so judgment is not required where proof is available. This determination extends that principle from a native protective function to a *constructed* verified element, under conditions that make the construction's decision equally bounded by proof.

3. The criterion: bounded by proof, in both parts

An authorization decision about a proposed action decomposes into two parts, and the criterion applies to **both**:

- **Admissibility** — is the action within a pre-authorized, verified region?
- **Selection** — which action, at what magnitude, in what order, at what rate, within that region?

A construction that bounds admissibility but leaves selection free **has not** produced an authorization decision bounded by proof, because the selection remains unbounded (FD-BL §6.1). Advance discharge is available only where a construction bounds **admissibility and selection together**. The conditions in §4 are the conditions under which that holds.

4. The determination

Advance discharge of the §4.5 authorization is permitted for the narrow class of above-line actions FD-BL §6.1 describes — those for which the authorization decision is expressible as a deterministic check on measured state, evaluated by a formally verified element the generative component cannot modify and cannot bypass — if and only if the construction satisfies all of R1, R2, and R3.

R1 — The whole safety case lives in a verified gatekeeper

The authorizing element is a deterministic, formally verified **gatekeeper** through which every above-line action in the class must pass, and which the generative **proposer** can neither modify nor bypass. The proposer holds **proposal authority only**; the gatekeeper holds **veto authority** and is the sole path by which any proposal reaches the process. Because the entire safety case lives in the gatekeeper, the proposer's non-determinism is irrelevant to safety: it may be opaque, probabilistic, or occasionally wrong, and still cannot cause an unsafe action, because every action it proposes is disposed of by a component that carries the safety case.

R2 — Completeness over state, trajectory, and rate (this is what bounds selection)

The gatekeeper's acceptance set shall be **proven** to be a subset of the truly-safe set over **state, trajectory, and timing/rate** — not a setpoint range check. Soundness requires that every admitted action is genuinely safe in the current state, on the current trajectory, at the current rate. Completeness requires that this holds across the whole domain the proposer can explore, and that **no admissible sequence, ordering, or rate of admissible actions can compose into an unsafe path** (anti-ratcheting and anti-oscillation). Only where this proof exists is **selection**, and not merely admissibility, bounded by

proof. **Where the proof cannot be produced, advance discharge is not available and the action takes in-the-moment Command Broker authorization under FD-BL §4.5.** An acceptance set shown only over static bounds is a range check and does not satisfy R2.

R3 — Envelope changes are categorically excluded from advance discharge

An action that changes the pre-authorized envelope itself — or the gatekeeper’s configuration, limits, or logic, or that moves the operating region toward a protective setpoint — is **not** eligible for advance discharge under any construction. No runtime gate can validate a change to its own safety basis. Such an action is an above-line action that requires either in-the-moment Command Broker authorization or an offline requalification that produces a new verified gatekeeper artifact before it is fielded. A live-learning element that updates the gatekeeper in the field is prohibited; any retrain is an offline event producing a new artifact that re-enters requalification.

4.4 What changes, and what does not

Under R1–R3 the human authorization is **not skipped; it is relocated in time** to the point at which a qualified human (a) defined and authorized the verified envelope, including any setback from protective setpoints; (b) accepted the completeness proof required by R2; and (c) retains sole authority over any change to the envelope under R3. Every runtime action in the class is still authorized — by a proof a qualified human accepted — exactly as a protective function’s actions are authorized in advance by its qualification. Placement, granularity, and scope are untouched (§2.2). Where R2 fails or R3 applies, the obligation is discharged in the moment, unchanged from FD-BL §4.5.

5. The two-object model (settling FD-BL §6.1’s explicit question)

FD-BL §6.1 asks whether the consequence classifier and the verified check are one object or two. **They are two distinct objects, in series, with distinct obligations, and shall not be merged.**

- The **consequence classifier** answers **placement** (FD-BL §4.2–§4.3): for each action, is it above or below the line? It routes action by action. It is not a safety gate and does not authorize.
- The **gatekeeper** answers **mode of discharge**: for an above-line action, is it inside a valid pre-authorized envelope satisfying R1–R3, such that its §4.5 authorization may discharge in advance? It vetoes; it does not classify.

Classification occurs first; the gatekeeper applies only to the above-line actions, and only where a valid envelope exists. Merging the two collapses a placement question into a discharge question and reintroduces the composite-classification error foreclosed at §6. A supplier shall implement them as two objects, and the evidentiary record shall show both the classifier’s routing basis and the gatekeeper’s acceptance-set proof.

6. Foreclosures retained

Two routes are foreclosed by FD-BL §6.1 and remain foreclosed; this determination does not reopen them:

- **Composite classification is foreclosed.** Classifying a system by the properties of the composite (generative component plus verified supervisor) rather than of the generating component is a change to placement — it makes architecture the measure of where the line falls, the same structural error as the technology-class and layer formulations. This determination changes mode, not placement; the action remains above the line, and the gatekeeper supplies the *authorization*, not a reclassification.

- **A third tier is foreclosed.** An intermediate classification with architectural entry criteria makes architecture determinative of placement. It is foreclosed on criterion-integrity grounds, overriding the assessment that it is the cleaner route: containing a change by defining a category on the wrong property preserves the defect it was meant to avoid.

7. Sector instantiation requirements

This determination is sector-neutral. Sector annexes shall instantiate it, and shall in particular specify:

- **Denied-condition immutability.** Under denied conditions the pre-authorized envelope shall become **immutable**, and re-engagement shall require a deliberate, authenticated act rather than a proposal. This determination requires the annex to specify the denied-condition behavior; it does not itself set qualification grade. For nuclear, the protection-grade qualification bar and the behavior under denial belong to the nuclear annex.
- **Materially-influenced reach preserved.** An action initiated or **materially influenced by** a generative component remains above the line (FD-BL §4.2; CB-Annex-Nuclear; DBA-ES). Advance discharge does not narrow that reach: a materially-influenced above-line action is eligible for advance discharge only through a gatekeeper satisfying R1–R3, and otherwise takes in-the-moment authorization.
- **Qualification of the envelope author.** The person who authors, reviews, and accepts a pre-authorized envelope, and who accepts the R2 completeness proof, shall be qualified for that function. This is distinct from the qualification to serve as an in-the-moment Command Broker; the qualification standard shall address both.

8. Accountability

Accountability for actions discharged in advance rests with the **named qualified human who authorized the envelope and accepted its proof, and with the asset owner-operator** — the same allocation FD-BL §4.5 makes for in-the-moment authorization. A supplier may build and verify the gatekeeper; a supplier may not hold the accountability, and no software component holds it. Any change to the envelope under R3 re-invokes this accountability at the point of requalification.

9. Relationship to adjacent open items

- **Indication integrity (FD-BL §6.2 / FD-BL-D2).** Resolved. A gatekeeper is bounded by proof only if its inputs are trustworthy; a gatekeeper acting on corrupted or drifted indication is not bounded by proof. The requirement that the measured state on which the gatekeeper’s deterministic check operates be traceable to independently verifiable values — stated on an interim basis in the v0.1 issue of this determination — is now settled generally by **FD-BL-D2 (indication integrity)**, which binds both this gatekeeper-input case and the human-broker case under one rule (traceability to independently verifiable values, and independent corroboration as a hard condition). Advance discharge under R1–R3 is valid only on inputs satisfying FD-BL-D2, I1–I3. See FD-BL-D2.
- **Terminology reservation (FD-BL §6.3 / FD-EV).** Resolved. This determination uses “pre-authorized envelope” for the proof-bounded region within which advance discharge is valid — the authorization sense (Sense G) now reserved by **FD-EV**. The operational sense of “envelope” (a region whose violation is a runtime event to detect) is renamed “operating band” by FD-EV; this determination’s usage is conforming and unchanged. See FD-EV.

10. Conformance and amendment set

This determination governs the mode of discharge of the FD-BL §4.5 obligation. Instruments carrying operative Command Broker requirements shall conform to it and shall **propagate by reference** (they cite FD-BL-D1; they do not restate it). The amendment set, carried in the Pass 2 conformance drafts:

- **CB Proposer–Gate Pattern** — cited as the architectural basis; its Condition I (completeness over state/trajectory/rate) and Condition II (action-within vs change-to the envelope) are made normative via R2 and R3.
- **Command Broker Implementation Guide; CB-IB Qualification Standard** — add the pre-authorization mode and the qualification of the envelope author (§7).
- **CB-CT Commissioning and Joint Training** — add the gate for authoring, reviewing, and approving a pre-authorized envelope and the accountable party for one; redirect the material-change test so that an envelope change routes to requalification rather than broker re-qualification.
- **ISA Parent Framework §6.2/§6.3; IEEE V&V §5.3** — align operator-as-executor and supplier-conformance language with the two-object model (§5) and R1–R3.
- **CB-Annex-Nuclear §3.2/§3.4** — retain the materially-influenced test; specify denied-condition immutability (§7).
- **DBA-W §5.5.4** — couple to §6.2 (gatekeeper inputs and indication integrity).
- **62443 AI Gap Analysis FR 2** — add, to the use-control taxonomy proposed to the standards committee, a category for proof-bounded pre-authorized autonomous operation, distinct from prohibited-autonomous and operator-guided.
- **DBA-GAS-FC2; DBA-ES-GN-FC8** — harvested as instantiation and evidence (the operating-envelope-with-setback; the firmware-trust precondition on the carve-out), not merely as recipients.

11. Open at this version

- Methods for demonstrating R2 completeness (state/trajectory/rate) tightly enough to be accepted rather than assumed, for high-dimensional proposers, are not standardized; this determination requires the proof but does not prescribe the method.
- The proposer–gatekeeper interface contract (what a proposal must contain; how vetoes are surfaced; how proposer liveness is reconciled with the gate’s safety obligation) is left to the Command Broker specification.
- Sector qualification bars for the envelope author and for denied-condition behavior are left to the sector annexes.

Revision History

0.3 — August 2026. Conformed the §9 terminology-reservation bullet to FD-EV: the envelope hold is discharged, the reserved “pre-authorized envelope” usage is confirmed conforming, and the operational sense’s rename to “operating band” is cited. No change to R1–R3, the two-object model, or the amendment set.

0.2 — August 2026. Conformed §9 to FD-BL-D2 (indication integrity): the interim indication-integrity floor is superseded and now points to FD-BL-D2, which settles the gatekeeper-input and human-broker cases under one rule. No change to R1–R3, the two-object model, the foreclosures, or the amendment set.

0.1 — August 2026. Initial issue, resolving FD-BL v0.2 §6.1. Determines that advance discharge of the §4.5 authorization is permitted for the narrow class, under R1 (whole safety case in a verified

gatekeeper), R2 (proven completeness over state, trajectory, and rate — bounding selection, not only admissibility), and R3 (envelope changes categorically excluded); states the two-object model (consequence classifier and gatekeeper in series); retains the composite-classification and third-tier foreclosures; requires denied-condition immutability at the sector annex; and sets the amendment set for Pass 2.